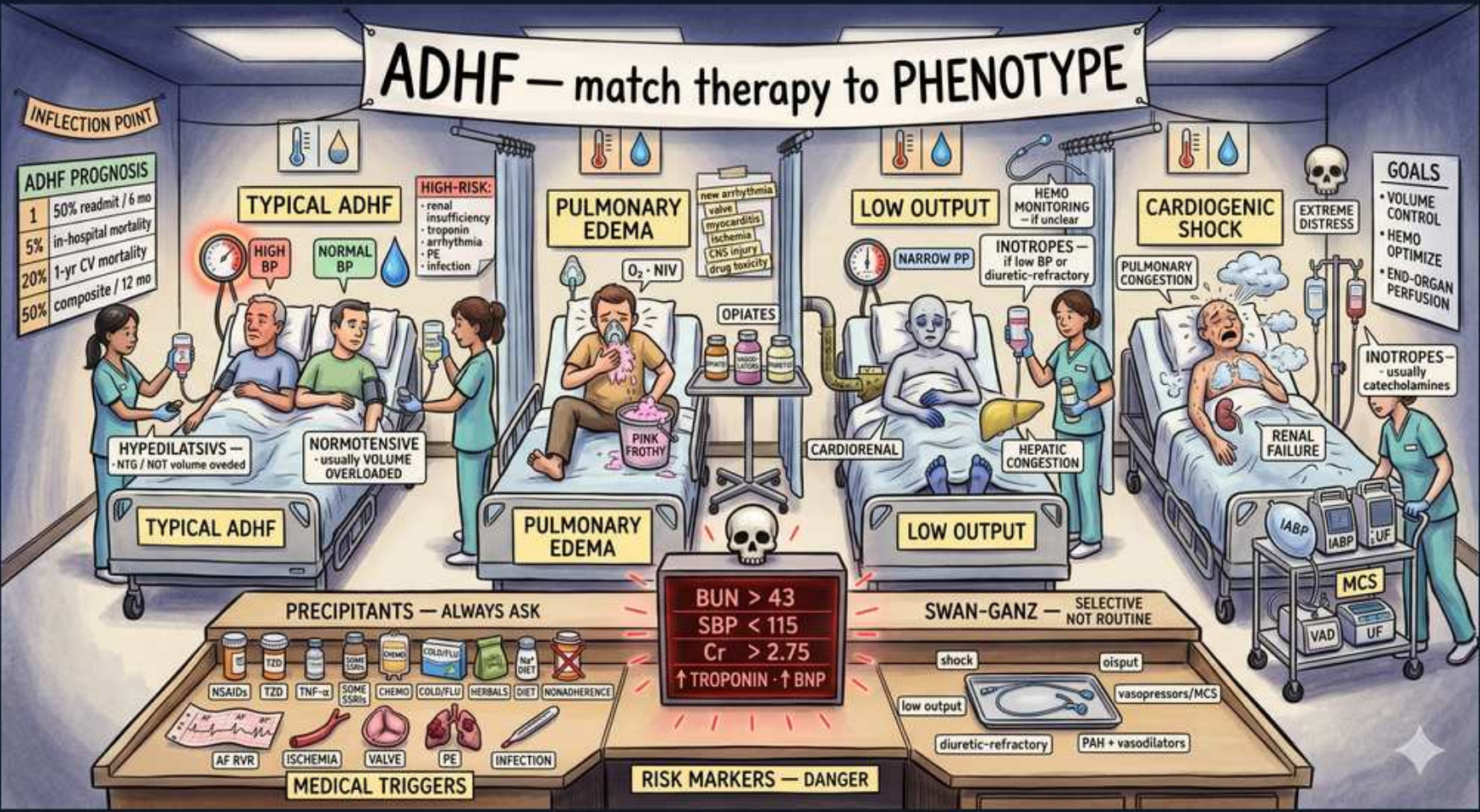


Heart Failure — SCENE 7: ADHF Phenotypes (ER Triage 4 Beds)



1. Typical ADHF — bed 1

WHAT YOU SEE

Bed 1, the 'TYPICAL ADHF' patient with a red HIGH-BP gauge above him — the hypertensive flash-edema phenotype (high pressure, redistribute not deplete).

WHAT IT ENCODES

Hypertensive ('vascular') ADHF is the most common ER presentation: rapid-onset 'flash' pulmonary edema driven by an abrupt rise in afterload and fluid REDISTRIBUTION into the lungs, NOT whole-body fluid accumulation — these patients are often warm-wet but near euvolemic in total body water. SBP is typically >140. Cornerstone therapy is IV vasodilators (nitroglycerin starting ~5–10 mcg/min titrated up, or nitroprusside) to drop preload/afterload; diuretics are secondary because there is little extra total-body volume to remove. Mnemonic: 'high pressure, redistribute not deplete.'

BOARD TRAP

WRONG answer: 'give high-dose IV furosemide as the primary treatment for a hypertensive flash-pulmonary-edema patient.' Discriminator: in hypertensive/vascular ADHF the problem is afterload-driven fluid shift, so aggressive diuresis of a near-euvolemic patient causes hypotension and AKI — the correct first move is a vasodilator (nitrates ± NIV).

2. Normotensive typical

WHAT YOU SEE

The 'NORMAL BP' gauge variant at the same first bed labelled 'usually volume overloaded' — the gradual cardiac-decompensation phenotype that is truly fluid-loaded.

WHAT IT ENCODES

Normotensive ADHF (SBP ~ normal) is the classic gradual 'cardiac' decompensation: days-to-weeks of progressive sodium/water retention with rising weight, edema, and elevated JVP — these patients ARE truly volume overloaded. IV loop diuretic is the cornerstone; per DOSE trial give an IV dose ~2.5× the chronic oral loop dose and target a falling JVP/weight. Distinguish from hypertensive ADHF where total body water is near normal.

BOARD TRAP

WRONG answer: 'treat every normotensive ADHF patient like flash edema with vasodilators only and minimal diuresis.' Discriminator: normotensive gradual-onset ADHF reflects true total-body volume overload — diuresis to euvolemia is the priority, unlike the redistribution physiology of hypertensive ADHF.

3. Pulmonary edema phenotype

WHAT YOU SEE

Bed 2, the 'PULMONARY EDEMA' patient in respiratory distress with an O2/NIV mask — severe hypoxic cardiogenic edema.

WHAT IT ENCODES

Acute cardiogenic pulmonary edema = severe respiratory distress with hypoxia, diffuse crackles, and bilateral 'bat-wing' infiltrates on CXR. Management bundle (mnemonic LMNOP-ish): sit Upright, high-flow O2/Noninvasive ventilation (CPAP/BiPAP), IV Nitrates for preload reduction, and diuretics if volume overloaded; morphine only cautiously. Target SpO2 ≥90%. NIV reduces intubation rates and work of breathing.

BOARD TRAP

WRONG answer: 'intubate immediately and start broad-spectrum antibiotics for the bilateral infiltrates.' Discriminator: bilateral symmetric perihilar edema with elevated JVP and S3 is cardiogenic — NIV plus nitrates usually averts intubation, and antibiotics treat pneumonia (asymmetric, fever, leukocytosis), not flash edema.

4. Opiates / NIV

WHAT YOU SEE

The 'OPIATES' tag and the 'O2 + NIV' mask at bed 2 — NIV is the workhorse, opiates are the discouraged option.

WHAT IT ENCODES

NIV (CPAP/BiPAP) is first-line adjunct in cardiogenic pulmonary edema: positive pressure recruits alveoli, lowers preload AND afterload, and reduces work of breathing and intubation rate. Opiates (low-dose IV morphine) modestly reduce preload and anxiety but are now DISCOURAGED for routine use — registry data link morphine to more ICU admissions, mechanical ventilation, and possibly higher mortality.

BOARD TRAP

WRONG answer: 'give IV morphine liberally to calm the dyspneic flash-edema patient.' Discriminator: routine morphine in ADHF is associated with respiratory depression, more intubations, and worse outcomes — NIV and nitrates are the evidence-based preload-reducing tools.

5. Low output phenotype

WHAT YOU SEE

Bed 3, the grey 'LOW OUTPUT' patient with a 'NARROW PP' (narrow pulse pressure) tag and cool/clammy look — the cold-wet hypoperfused phenotype.

WHAT IT ENCODES

Low-output ('cold') ADHF = inadequate cardiac output with hypoperfusion: cool/clammy extremities, NARROW pulse pressure (proportional pulse pressure <25%), altered mentation, rising creatinine, and congested liver. This is the 'cold-wet' Stevenson quadrant (low perfusion + congestion). Confirm hemodynamics; in unclear cases or shock a PA catheter can guide. Cardiorenal and cardiohepatic dysfunction are markers of severity.

BOARD TRAP

WRONG answer: 'the cool extremities and low urine output mean the patient is dehydrated — give a fluid bolus.' Discriminator: in cold-wet ADHF hypoperfusion is from pump failure WITH congestion, not hypovolemia — fluids worsen pulmonary edema; the patient needs decongestion ± inotropic support, not volume.

6. Inotropes if low BP / diuretic-refractory

WHAT YOU SEE

The 'INOTROPES — if low BP / diuretic-refractory' tag at bed 3 — the caption itself spells out the only two indications.

WHAT IT ENCODES

Inotropes (dobutamine ~2–10 mcg/kg/min, or milrinone 0.125–0.75 mcg/kg/min) are reserved for the cold/low-output patient with hypotension causing end-organ hypoperfusion OR genuine diuretic refractoriness from poor renal perfusion. They improve contractility and forward flow as a bridge, but increase myocardial O2 demand and arrhythmias and do NOT improve survival. Milrinone is renally cleared and a pulmonary vasodilator (good for RV failure/pulmonary HTN); avoid milrinone in profound hypotension.

BOARD TRAP

WRONG answer: 'add an inotrope to any ADHF patient who is still congested.' Discriminator: inotropes are only for low-output/hypoperfusion or diuretic-refractory states — in a warm-wet patient with adequate BP they add arrhythmia and mortality risk with no decongestion benefit (OPTIME-CHF showed harm with routine milrinone).

7. Cardiogenic shock — bed 4

WHAT YOU SEE

Bed 4, the 'CARDIOGENIC SHOCK' patient in 'EXTREME DISTRESS' with a 'RENAL FAILURE' tag — sustained hypotension plus end-organ failure.

WHAT IT ENCODES

Cardiogenic shock = sustained hypotension (SBP <90 for >30 min or pressors needed) PLUS end-organ hypoperfusion (lactate >2, oliguria, cold mottled skin, altered mentation) with cardiac index typically <2.2 L/min/m² and PCWP >15. Mortality ~40–50%. Find and fix the cause — ACS is #1, so emergent revascularization (PCI) per the CULPRIT-SHOCK trial: revascularize the culprit lesion ONLY, not all vessels.

BOARD TRAP

WRONG answer: 'in MI-related cardiogenic shock, perform immediate complete multivessel revascularization of all stenoses.' Discriminator: CULPRIT-SHOCK showed culprit-lesion-only PCI lowered 30-day death/renal failure vs immediate multivessel PCI — stage non-culprit lesions later.

8. Inotropes/pressors + MCS

WHAT YOU SEE

The 'INOTROPES — mostly catecholamines' note and the 'IABP/VAD' (MCS) device beside bed 4 — norepinephrine first, mechanical support as selective rescue.

WHAT IT ENCODES

Cardiogenic shock support: vasopressor of choice is NOREPINEPHRINE (SOAP-II — dopamine caused more arrhythmias and trended to higher mortality), plus an inotrope (dobutamine) for contractility. Add mechanical circulatory support — IABP, Impella (pVAD), or VA-ECMO — for refractory shock as a bridge to recovery/transplant. Note: the IABP-SHOCK II trial showed routine IABP did NOT reduce mortality in MI shock.

BOARD TRAP

WRONG answer: 'start dopamine as the first-line vasopressor and place a routine IABP to improve survival in MI cardiogenic shock.' Discriminator: norepinephrine beats dopamine (SOAP-II) and routine IABP gave no mortality benefit (IABP-SHOCK II) — MCS is selective, not reflexive.

9. Precipitants — always ask

WHAT YOU SEE

The bottom 'PRECIPITANTS — ALWAYS ASK' shelf of trigger icons (pill bottles, AF heart, ischemic heart, lungs/PE, infection) — the mandatory hunt for what tipped the patient over.

WHAT IT ENCODES

Every ADHF admission demands a hunt for the precipitant because treating it changes outcomes. Mnemonic FAILURE / 'CHAMP': Forgot meds/Faild compliance, Arrhythmia (esp new AF with RVR), Ischemia/ACS, Lifestyle (dietary salt/fluid, NSAIDs, alcohol), Upregulation (thyrotoxicosis, pregnancy, anemia), Rheumatic/valve disease, Embolism (PE), plus infection/sepsis. CHAMP = Coronary, Hypertension emergency, Arrhythmia, acute Mechanical (papillary rupture, VSD), PE.

BOARD TRAP

WRONG answer: 'just treat the congestion with diuretics and discharge — the trigger does not matter.' Discriminator: missing AF with RVR, ACS, or a new severe valve lesion leads to early readmission or death; identifying/correcting the precipitant is a board-mandated step in ADHF workup.

10. BUN > 43

WHAT YOU SEE

The red 'BUN > 43' marker on the central 'RISK MARKERS — DANGER' stop-sign board — the top node of the ADHERE in-hospital mortality tree.

WHAT IT ENCODES

BUN >43 mg/dL is the single STRONGEST predictor of in-hospital mortality in the ADHERE registry risk tree — it reflects both renal hypoperfusion and intense neurohormonal (RAAS/ADH) activation. The ADHERE decision tree splits first on BUN ≥43, then SBP <115, then creatinine ≥2.75; the highest-risk branch (all three) carries ~22% in-hospital mortality vs ~2% for none.

BOARD TRAP

WRONG answer: 'an elevated BUN in ADHF is just dehydration and is prognostically meaningless.' Discriminator: BUN >43 is the top node of the ADHERE mortality tree — it is a high-risk marker reflecting neurohormonal activation/hypoperfusion, not a benign lab.

11. SBP < 115

WHAT YOU SEE

The 'SBP < 115' line on the red 'RISK MARKERS — DANGER' board — the second ADHERE splitting node; low admission pressure signals low output.

WHAT IT ENCODES

Admission SBP <115 mmHg is the SECOND node of the ADHERE risk tree and a powerful adverse prognostic marker — LOW blood pressure in ADHF signals a low-output, poorly perfusing state, the opposite of the better-prognosis hypertensive phenotype. Higher admission SBP actually correlates with LOWER mortality in ADHF.

BOARD TRAP

WRONG answer: 'a low SBP on ADHF admission is reassuring because the patient is not hypertensive.' Discriminator: low admission SBP marks low cardiac output and predicts worse outcomes; the hypertensive (high-SBP) phenotype carries the better prognosis.

12. Cr > 2.75 / rising troponin

WHAT YOU SEE

The 'Cr > 2.75' line plus '↑TROPONIN ↑BNP' on the red 'RISK MARKERS' board — the third ADHERE node and the prognostic biomarker flags.

WHAT IT ENCODES

Creatinine ≥2.75 mg/dL is the third splitting node of the ADHERE tree (cardiorenal injury). A detectable/rising troponin in ADHF — even without overt ACS — independently predicts higher mortality (ongoing myocyte injury), and a rising BNP/NT-proBNP trend tracks worsening wall stress. Together these mark the highest-risk patients.

BOARD TRAP

WRONG answer: 'a detectable troponin in an ADHF patient without chest pain is always a STEMI requiring emergent cath.' Discriminator: low-level troponin elevation is common in ADHF from demand ischemia/wall stress and is prognostic — distinguish a flat low-level elevation from a dynamic rise-and-fall pattern that signals true type 1 MI.

13. Swan-Ganz — selective only

WHAT YOU SEE

The 'SWAN-GANZ — SELECTIVE NOT ROUTINE' PA catheter on the right counter — reserved for shock/refractory cases, not every admission (ESCAPE).

WHAT IT ENCODES

The pulmonary artery (Swan-Ganz) catheter is NOT used routinely: the ESCAPE trial showed no improvement in mortality or days alive out of hospital, with more catheter-related adverse events. Reserve it for selected cases — cardiogenic shock with unclear hemodynamics, diuretic-refractory congestion, suspected mixed/discordant shock, or evaluation before transplant/MCS — where it guides tailored therapy (measures PCWP, CO, SVR).

BOARD TRAP

WRONG answer: 'place a PA catheter routinely in all ADHF admissions to optimize fluid management and improve outcomes.' Discriminator: ESCAPE proved routine PA catheterization gives no outcome benefit and adds harm — it is a selective tool for shock or refractory/uncertain hemodynamics only.
